

AAD Midsem Answer Key

Problem 1

1. The FFT of $(1, 0, 1, -1)$

Explanation: For a 4-point sequence $x[n] = \{x_0, x_1, x_2, x_3\}$, the DFT $X[k]$ is calculated as:

$$X[k] = \sum_{n=0}^3 x[n] \cdot e^{-j2\pi nk/4}$$

Here, $x = \{1, 0, 1, -1\}$. The twiddle factor is $W_4 = e^{-j\pi/2} = -j$.

- $X[0] = x[0] + x[1] + x[2] + x[3] = 1 + 0 + 1 - 1 = 1$
- $X[1] = x[0]W_4^0 + x[1]W_4^1 + x[2]W_4^2 + x[3]W_4^3 = 1(1) + 0(-j) + 1(-1) + (-1)(j) = 1 - 1 - j = -j$
- $X[2] = x[0]W_4^0 + x[1]W_4^2 + x[2]W_4^4 + x[3]W_4^6 = 1(1) + 0(-1) + 1(1) + (-1)(-1) = 1 + 1 + 1 = 3$
- $X[3] = x[0]W_4^0 + x[1]W_4^3 + x[2]W_4^6 + x[3]W_4^9 = 1(1) + 0(j) + 1(-1) + (-1)(-j) = 1 - 1 + j = j$

Final Answer: $(1, -j, 3, j)$

2. The worst-case complexity of deterministic quick-sort

Explanation: The worst-case for quick-sort occurs when the pivot selection consistently leads to the most unbalanced partition. In a deterministic implementation (ie. always choosing the first or last element), this happens if the array is already sorted. Each partition step divides the array of size n into subarrays of size 0 and $n - 1$. The recurrence relation becomes:

$$T(n) = T(n - 1) + \Theta(n)$$

This recurrence expands to an arithmetic series, resulting in a quadratic time complexity.

Final Answer: $O(n^2)$

3. The g.c.d. of two consecutive Fibonacci numbers

Explanation: Let the numbers be F_{n+1} and F_n . By the Euclidean algorithm, $\gcd(a, b) = \gcd(b, a \pmod{b})$. Since $F_{n+1} = F_n + F_{n-1}$, we have:

$$\gcd(F_{n+1}, F_n) = \gcd(F_n, F_{n+1} - F_n) = \gcd(F_n, F_{n-1})$$

Repeating this process gives $\gcd(F_2, F_1) = \gcd(1, 1) = 1$.

Final Answer: 1

4. Solving the recurrence $T(n) = O(n) + \sum_{i=2}^{\lfloor \sqrt{n} \rfloor} T(\lfloor \frac{n}{i^2} \rfloor)$, $T(\leq 1) = 1$

Final Answer: $O(n)$

5. **A Huffman code for aababcbabcdabcdeabcdef**

Explanation: Huffman coding uses character frequencies to build an optimal prefix code.

- (a) Frequencies: a:6, b:5, c:4, d:3, e:2, f:1
- (b) Build Tree: Greedily combine the two nodes with the lowest frequencies. One possible tree structure yields the following codes by assigning '0' to left branches and '1' to right branches.

Final Answer: {a:10, b:01, c:00, d:110, e:1111, f:1110} (Note: Other valid codes are possible).

6. **It is possible to find the median of n numbers in $O(\text{---})$ time.**

Explanation: The median can be found in linear time using the "Median of Medians" algorithm. This algorithm provides a deterministic worst-case linear-time performance for the selection problem, which includes finding the median.

Final Answer: $O(n)$

7. **The edit-distance between SUNNY and SNOWY**

Explanation:

- (a) 'SUNNY' $\rightarrow$ 'SNNNY' (Substitute U $\rightarrow$ N)
- (b) 'SNNNY' $\rightarrow$ 'SNONY' (Substitute N $\rightarrow$ O)
- (c) 'SNONY' $\rightarrow$ 'SNOWY' (Substitute N $\rightarrow$ W)

Final Answer: 3

8. **An optimal parenthesization of a matrix-chain product with sequence of dimensions {5, 10, 3, 12, 5, 50, 6}**

Explanation: Compute the minimum cost $m[i, j]$ for multiplying matrices $A_i \dots A_j$. By filling out the cost table and a corresponding split table $s[i, j]$, we find the optimal cost is 2010. The s table is then used to reconstruct the parenthesization. The final split for $A_1 \dots A_6$ is at $k = 2$, giving $(A_1 A_2)(A_3 \dots A_6)$. The split for $A_3 \dots A_6$ is at $k = 4$, giving $(A_3 A_4)(A_5 A_6)$.

Final Answer: $((A_1 A_2)((A_3 A_4)(A_5 A_6)))$

9. **The cut-property states that**

Final Answer: for any cut of a graph, a minimum-weight edge crossing the cut belongs to some Minimum Spanning Tree.

10. **A language $A \subset \{0, 1\}^*$ is said to be mapping-reduced to language $B \subset \{0, 1\}^*$ if there exists a computable function f such that for all inputs $w \in \{0, 1\}^*$,**

Explanation: The computable function f must transform any instance w of problem A into an instance $f(w)$ of problem B such that the answer (yes/no, or member/non-member) is preserved.

Final Answer: $w \in A \iff f(w) \in B$